

Task 1: Business Requirements Analysis

Ubuntu Innovations (Pty) Ltd

Week 1 – IT Support Capstone Project

Prepared for: Group Capstone Project

1. Company Overview

Ubuntu Innovations (Pty) Ltd is a rapidly growing technology startup based in Cape Town, South Africa. The company is relocating to a new office and requires a fully documented IT support and infrastructure solution.

The company has 25 employees working across five departments. As a technology-focused organisation, Ubuntu Innovations requires reliable, secure and efficient IT infrastructure to support its daily business activities.

The main business requirements include secure wired and wireless connectivity, shared file storage, shared printers, employee user accounts, departmental access control, antivirus and endpoint security, data backup and recovery, security policies, troubleshooting procedures and documentation for future IT staff.

2. Number of Users and Departments

Ubuntu Innovations has 25 employees divided across five departments:

Department	Number of Employees
Executive Management	3
Finance	4
Human Resources	3
Sales and Marketing	7
Software Development	8
Total	25

Each department requires appropriate access to company systems and information. Access should be controlled according to employees' departmental responsibilities. This supports the requirement for access control based on departments.

3. Business Applications Required

The organisation requires applications that support general office work, communication, file management, security and department-specific activities.

Department	Required Applications	Purpose
Executive Management	Office/productivity software, email, web browser, communication tools	Management, communication, reporting and decision-making
Finance	Spreadsheet/accounting software, email, office applications	Financial records, calculations, reporting and administration
Human Resources	Word processing, spreadsheets, email and document management	Employee records, documentation and communication

Department	Required Applications	Purpose
Sales & Marketing	Office applications, email, web browser and communication tools	Customer communication, marketing activities and reporting
Software Development	Development tools, code editor/IDE, web browser, version control	Software development, testing and project work
All Departments	Web browser, email, productivity software, antivirus/endpoint protection	Daily operations, communication and device protection

The organisation also requires shared file storage and shared printers. Ubuntu Linux is identified in the project as an operating system for practice, while GitHub is recommended for version control and submission.

4. Data Security Requirements

The following security requirements should be implemented:

- **User authentication:** Every employee should have an individual user account rather than sharing accounts.
- **Departmental permissions:** Users should only receive access to files and resources necessary for their jobs.
- **Antivirus and endpoint protection:** Company computers should have appropriate protection against malware and other threats.
- **Data backup:** Important business information should be backed up regularly to support recovery.
- **Password and authentication security:** Security controls should include password complexity and multi-factor authentication.
- **Device protection:** Device encryption and patch management should be included as part of the security approach.
- **Access control:** Sensitive departmental information should only be accessible to authorised users.

5. Connectivity Needs

Ubuntu Innovations requires both wired and wireless network connectivity to support its 25 employees and shared resources.

Network Requirement	Proposed Solution
Internet connectivity	Business-grade firewall router
Wired network	24-port Gigabit managed switch
Wireless network	2 × Dual-band Wi-Fi 6 access points
Shared printing	2 network laser printers
File storage	8TB RAID NAS device
Power protection	2 × 1500VA UPS units

The network should provide reliable connectivity while protecting company resources. The project later proposes separate subnets for servers, user devices, printers and Wi-Fi to support organised network management.

6. Operational Challenges

6.1 Office relocation: The company is moving to a new office, so the IT infrastructure needs to be planned and documented to minimise disruption.

6.2 Reliable connectivity: Network or internet outages could affect communication, shared resources and employee productivity.

6.3 Data access and permissions: Different departments work with different information, making appropriate permissions necessary.

6.4 Data loss: Important company files need protection against accidental deletion, hardware failure and other incidents through a backup and recovery strategy.

6.5 Cybersecurity threats: The organisation needs protection against risks such as phishing and ransomware, as well as hardware and power-related risks.

6.6 Hardware and technical problems: Common support issues may include computers not powering on, internet connectivity problems and printers going offline.

6.7 Future IT support: Systems must be properly documented so future IT staff can understand the environment and troubleshoot problems efficiently.

7. Summary of Business Requirements

Area	Requirement
Company	Ubuntu Innovations (Pty) Ltd, Cape Town
Employees	25
Departments	5
Connectivity	Secure wired and wireless network
Internet	Reliable business internet connectivity
File Storage	Shared company storage
Printing	Shared network printers
User Accounts	Individual employee accounts
Access Control	Department-based permissions
Security	Antivirus and endpoint security
Backup	Regular data backup and recovery
Documentation	IT support and troubleshooting documentation
Network Equipment	Router/firewall, managed switch and Wi-Fi access points
Storage Equipment	NAS with RAID
Power Protection	UPS units

8. Conclusion

The business requirements analysis shows that Ubuntu Innovations (Pty) Ltd requires a secure, reliable and well-organised IT infrastructure to support its 25 employees across five departments.

The key requirements are reliable wired and wireless connectivity, appropriate business applications, individual user accounts, departmental access controls, endpoint security, shared file storage, shared printing, data backup and recovery, and proper IT documentation.

These requirements provide the foundation for the next stages of the capstone project, including the hardware inventory, software inventory, network diagram and troubleshooting documentation.

Source: Four Week Capstone Project Guide – Design and Implementation of an IT Infrastructure and Support Solution for Ubuntu Innovations (Pty) Ltd.